Lecture 8 Second Order Homogeneous ODEs

Lecture 8
Second O...

Engineering Mathematics III

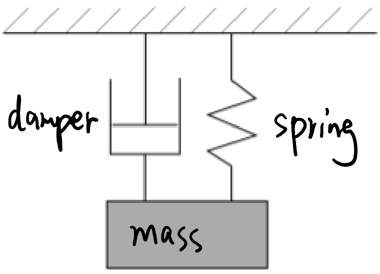
ENGE702

Lecture 8: Second Order Homogeneous ODEs

Second order ODEs

Second order linear homogeneous ODEs with constant coefficients have a number of fundamental physical models. We will look at one called a **Mass - Spring - Damper System**.

This consists of a mass that is attached to a fixed structure via a spring and a damper, as in the following schematic:



Second order ODEs

We need to set up a model to describe the position of the mass at any time.

We will take the equilibrium position of the mass to be $y = 0$, and downwards to be positive displacement.

We will look at the forces acting on the mass.

Hooke's law says that the force from a spring is proportional to its displacement from its equilibrium position, and acts in the opposite direction to the displacement. Thus the force from the spring can be given by

spring constant

$$F_s = -ky \rightarrow \text{displacement}$$

The force from a damper is proportional to the velocity of the motion (again in the opposite direction to the velocity). Thus the force from the damper can be given by

damper constant

$$F_d = -cy' \rightarrow \text{velocity}$$

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Newton's second law tells that

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Finally, Newtons law says that the sum of the forces acting on the mass is equal to the mass times acceleration, so

$$F = my''$$

Here F is the sum of all forces, so $F = F_s + F_d$, or

$$my'' = -cy' - ky,$$

rearranging we get

$$my'' + cy' + ky = 0.$$

This is a second order linear homogeneous ODE with constant coefficients, and we will look at the solutions for various values of m, c, and k.

Note in our model that $m, k > 0$ and $c \geq 0$.

Newton's second law tells that

$$F = m\underline{y''} \quad \text{acceleration}$$

$$\Rightarrow -ky - cy' = my''$$

$$\Rightarrow m\underline{y''} + c\underline{y'} + k\underline{y} = 0$$

second order linear homogeneous ODE

$$my'' + cy' + ky = 0$$

Case 1: no damping, so the damping constant $c = 0$.

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Case 1: no damping.

In this case our differential equation is

$$my'' + ky = 0$$

Since $m, k > 0$, we can put this in standard form and solve the characteristic equation $\lambda^2 + \frac{k}{m} = 0$

This has solutions

$$\lambda = \pm\sqrt{\frac{k}{m}}i$$

so the system has angular frequency

$$\omega = \sqrt{\frac{k}{m}}$$

Since $-\frac{a}{2} = 0$, the general solution will be

$$y = A\cos(\omega t) + B\sin(\omega t)$$

displacement as a function of time.

$$my'' + ky = 0$$

characteristic equation: $m\lambda^2 + k = 0$

$$\Rightarrow \lambda^2 + \frac{k}{m} = 0$$

$a = 1$, $b = 0$, $c = \frac{k}{m}$

quadratic formula

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{0 \pm \sqrt{0^2 - 4 \times \frac{k}{m}}}{2}$$

$$= \frac{\pm 2\sqrt{-\frac{k}{m}}}{2} = \pm\sqrt{\frac{k}{m}}\sqrt{-1} = 0 \pm \sqrt{\frac{k}{m}}\, i.$$

The solution to ODE is

$$y(t) = e^{0t}\left(A\cos\left(\sqrt{\frac{k}{m}}t\right) + B\sin\left(\sqrt{\frac{k}{m}}t\right)\right)$$

$$= A\cos(\omega t) + B\sin(\omega t), \quad \omega = \sqrt{\frac{k}{m}}.$$

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Example: Suppose we have a mass - spring system with mass of 4 and spring constant of 16. The displacement at $t = 0$ is 1 and the velocity at $t = 0$ is 2. Find a function that describes the position as a function of time.

$m = 4$, $k = 16$, $c = 0$

The ODE is $my'' + ky = 0$

$$4y'' + 16y = 0 \Rightarrow y'' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 2.$$

$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{16}{4}} = 2$, so the general solution is $y(t) = A\cos(2t) + B\sin(2t)$

$$\Rightarrow y'(t) = -2A\sin(2t) + 2B\cos(2t)$$

Apply the initial conditions: $y(0) = A\cos(0) + B\sin(0) = A = 1$

$$y'(0) = -2A\sin(0) + 2B\cos(0) = 2B = 2 \Rightarrow B = 1$$

The particular solution is $y(t) = \cos(2t) + \sin(2t)$.

Second order ODEs

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Second order ODEs

Example: Suppose we have a mass - spring system with mass of 4 and spring constant of 16. The displacement at $t = 0$ is 1 and the velocity at $t = 0$ is 2. Find a function that describes the position as a function of time.

First we know that $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{16}{4}} = 2$. So our general solution is

$$y = A\cos(2t) + B\sin(2t).$$

Differentiating this gives

$$y' = -2A\sin(2t) + 2B\cos(2t).$$

Our initial conditions give $y(0) = 1$ and $y'(0) = 2$.

So we have

$$\begin{aligned} 1 &= A\cos(0) + B\sin(0) \\ 2 &= -2A\sin(0) + 2B\cos(0) \end{aligned}$$

$y'(0) = -2A\sin(0) + 2B\cos(0) = 2B = 2 \Rightarrow B = 1$

The particular solution is $y(t) = \cos(2t) + \sin(2t)$.

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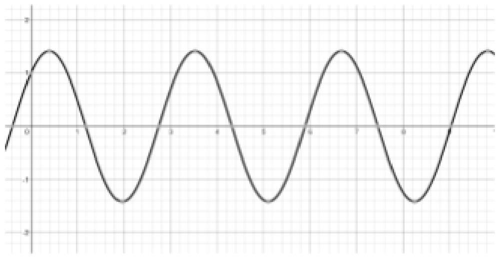
Solving this gives $A = 1$ and $B = 1$, so our solution is

$$y = \cos(2t) + \sin(2t).$$

Plotting this gives:

no damping, the mass oscillate around the equilibrium.



Changing initial conditions will alter the amplitude and phase, but the frequency ω will always stay the same.

Second order ODEs

Now lets see what happens when we add a damper to the system. Our differential equation is now:

$$my'' + cy' + ky = 0.$$

As we saw earlier, the solution type will vary depending on the roots of the characteristic equation.

The roots of the characteristic equation come from the quadratic formula, and are give by

$$\lambda_1 = -\alpha + \beta \quad \text{and} \quad \lambda_2 = -\alpha - \beta$$

where

$$\alpha = \frac{c}{2m} \quad \text{and} \quad \beta = \frac{1}{2m}\sqrt{c^2 - 4mk}$$

Consider a damper in the system.

$$my'' + cy' + ky = 0$$

Characteristic equation:

$$m\lambda^2 + c\lambda + k = 0$$

use the quadratic formula

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4km}}{2m} = -\underbrace{\boxed{\frac{c}{2m}}}_{\alpha} \pm \underbrace{\boxed{\frac{\sqrt{c^2-4km}}{2m}}}_{\beta} = -\alpha \pm \beta$$

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$$\alpha = \frac{c}{2m} \quad \beta = \frac{1}{2m}\sqrt{c^2 - 4mk}$$

There are three cases:

Case 1: Overdamping. In this case $c^2 > 4mk$, and there are two distinct real roots.

Case 2: Critical damping. In this case $c^2 = 4mk$, and there is one real root.

Case 3: Underdamping. In this case $c^2 < 4mk$, and there are complex

Second order ODEs

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Case 2: Critical damping. In this case $c^2 = 4mk$, and there is one real root.

Case 3: Underdamping. In this case $c^2 < 4mk$, and there are complex conjugate roots.

We will look at the general solutions of the three cases and give some particular solutions.

$= -\alpha \pm \beta$

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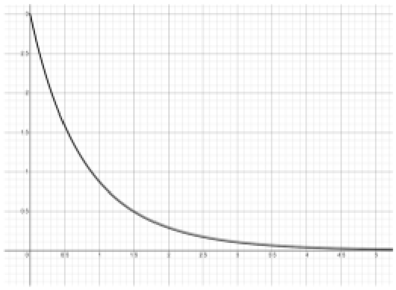
Case 1: Overdamping. Here the general solution is

$$y = c_1 e^{-(\alpha-\beta)t} + c_2 e^{-(\alpha+\beta)t}$$

Note that both exponents are negative due to the nature of α and β, so the function will tend to 0 as time increases.

A typical solution looks something like:



Case 1 over damping.

there exists two real roots

$\lambda_1 = -\alpha+\beta$, $\lambda_2 = -\alpha-\beta$

So the general solution is

$y(t) = C_1 e^{(-\alpha+\beta)t} + C_2 e^{(-\alpha-\beta)t}$.

After some time, the mass tends to the equilibrium without oscillation.

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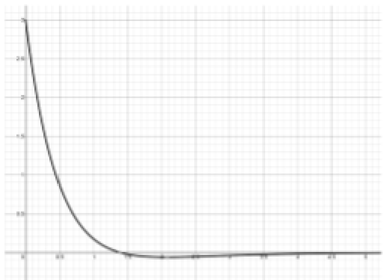
It is still possible that the mass passes equilibrium in this case, if the initial velocity is large enough, for example:



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Case 2: Critical damping. Here the general solution is

$$y = (c_1 + c_2 t)e^{-(\alpha t)}.$$

This is the 'borderline' case between oscillatory and non-oscillatory

Case 2: critical damping

$c^2 = 4mk \Rightarrow \beta = 0$

So only one real root $\lambda = -\alpha$.

And the general solution is

$y = (c_1 + c_2 t)e^{-\alpha t}$

This is the 'borderline' case between oscillatory and non-oscillatory motion. The solution looks much like the overdamped case graphically. It can be though of as getting to equilibrium 'as fast as possible'.

And the general solution is

$$y(t) = c_1 \cdot e^{-\alpha t} + c_2 \cdot t e^{-\alpha t}.$$

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Case 3: Underdamping.

This is the most complicated case. Here our solutions are

$$\lambda_1 = -\alpha + i\omega \quad \text{and} \quad \lambda_2 = -\alpha - i\omega,$$

and the general solution is:

$$y = e^{-\alpha t}\left(A\cos(\omega t) + B\sin(\omega t)\right).$$

The sin and cos terms mean the solution will oscillate about the equilibrium, but the $e^{-\alpha t}$ term means it will tend to the equilibrium as time increases (remember $\alpha = \frac{c}{2m}$ is always positive)

Case 3: underdamping

$c^2 < 4mk \Rightarrow$ two complex roots:

$\lambda_1 = -\alpha + i\omega \quad \lambda_2 = -\alpha - i\omega$

The general solution is

$$y(t) = e^{-\alpha t}\left(A\cos(\omega t) + B\sin(\omega t)\right)$$

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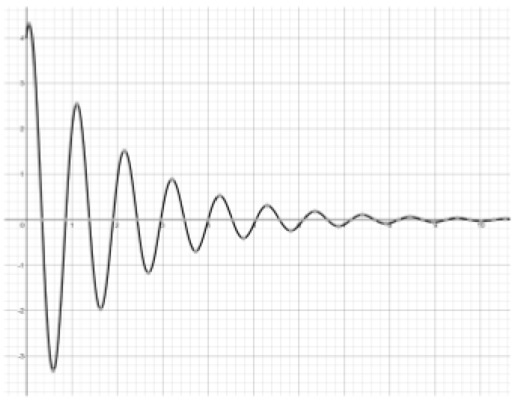
The solution will look something line this:



The mass will oscillate towards the equilibrium over time.

Exercises

Suppose we have a mass - spring - damper system with a mass of $m = 1$, and a damping constant of $c = 4$. Find a value of the spring constant k that will give:

- underdamped motion
- critically damped motion
- overdamped motion

Text Book References

Chapter 2.4